



CMA7 Motor Drive Unit

Operating Manual

HM 0.460.302 - 11/3/2025

Supported product model: CMA7



Important Note!

Read carefully before use.

Keep for future reference.

Legal Notice

Read this manual carefully before operating the equipment. Failure to follow the instructions may result in injury or damage. Shanghai Huaming Power Equipment Co., Ltd. (hereinafter referred to as "Huaming") shall not be liable for any losses or damages resulting from the use of the product in a manner inconsistent with this manual, except where such liability arises under applicable product liability laws.

Keep this manual for future reference.

The information in this document is based on the technical specifications at the time of printing. Product specifications are subject to change without notice as part of our continuous improvement efforts. Significant revisions will be incorporated into new editions of this manual. For the latest version, refer to our website or contact us.

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1. About This Manual

This manual provides an overview and essential safety instructions for the CMA7 Motor Drive Unit (CMA7). It also offers comprehensive instructions on transporting, installing, commissioning, and maintaining the CMA7 to ensure proper integration with the corresponding tap-changer and transformer.

1.1. Target audience

This manual is intended for qualified personnel involved in the installation, commissioning, operation, and maintenance of the CMA7.

Roles	Responsibilities
Technical Field Assistant (TFA) or Commissioning Engineer	• Installing and commissioning CMA7 • Integrating CMA7 into power transformers • Verifying functional and safety compliance
Operator	• Monitoring operating parameters • Performing routine tap-change operation • Reporting abnormal conditions
Maintenance Technician	• Performing inspection and maintenance • Conducting mechanical/electrical safety checks • Conducting minor repair operations



Important:

This equipment must only be operated by qualified technical personnel. Untrained personnel shall not operate, maintain, or modify CMA7. For required personnel qualifications, refer to [2.5 Personnel qualifications](#).

1.2. Document retention

Keep this manual and all related documents in a safe place for future reference.

1.3. Related documents

This manual should be used in conjunction with the following documents. Reading this manual in isolation may not provide a complete understanding of the product.

- Wiring Diagram
- Order Documentation
- Routine Test Report
- Packing list
- Tap-Changer manual
- Protective Relay manual
- Supplementary Documents (if any)

1.4. Conventions

1.4.1. Warning messages

The warning messages in this manual follow the format below:

⚠ DANGER	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
⚠ WARNING	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
⚠ CAUTION	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
NOTICE	
Identify type of risk Identify consequences of not avoiding the risk ✓ Identify how to avoid the risk	

Table 1-1 Classification of signal words

Signal word	Hazard level	Potential consequences
 Danger	Imminent hazard	Death or serious injury
 Warning	Potential hazard	Death or serious injury

Table 1-1 Classification of signal words (continued)

Signal word	Hazard level	Potential consequences
 Caution	Potential hazard	Minor or moderate injury
 Notice	Potential risks	Damage to the supported product (or to other property)

Table 1-2 Safety symbols

Symbol	Meaning
	Danger: Indicates a hazard that will result in serious injury or death if not avoided.
	Warning: Indicates a hazard that could result in serious injury or death if not avoided.
	Overhead load
	High voltage
	Explosive material
	Flammable material

1.4.2. Notes

The following symbols are used to highlight information worth noting:



Important:

Indicates key information and requires attention.



Note:

Indicates an ordinary note.

1.4.3. Procedural instructions

Procedural instructions are presented as single-step or multi-step operations.

Single-step operations

Pre-operation preparation (Optional)

Objective:

This is a step.

1. This is a substep.
2. This is a substep.

Specific instructions/Expected results (Optional)

Outcome (Optional)

Multi-step operations

Pre-operation preparation (Optional)

Objective:

1. This is a step.
2. This is a step.
 - a. This is a substep.
 - b. This is a substep.
 - c. This is a substep.

Specific instructions/Expected results (Optional)

Outcome (Optional)

1.5. Abbreviations

The following table lists the abbreviations used in this manual and their full forms.

Table 1-3 Abbreviations and full forms

Abbreviations	Full form
FAT	Factory Acceptance Test
MDU	Motor Drive Unit
OLTC	On-Load Tap-Changer
PPE	Personal Protective Equipment
TFA	Technical Field Assistant

2. Safety Notes

This manual is an integral part of the product.

Read this manual thoroughly to familiarize yourself with the product.

Pay close attention to all warning and cautionary information in this manual to prevent hazards.

Read and follow all safety instructions in this chapter.

While this product is designed and manufactured using state-of-the-art technology, improper use may result in personal injury, equipment damage, or property loss.

2.1. Safety instructions

To prevent possible injury, read and follow the instructions in the table below.

Table 2-1 Potential danger and preventive measures

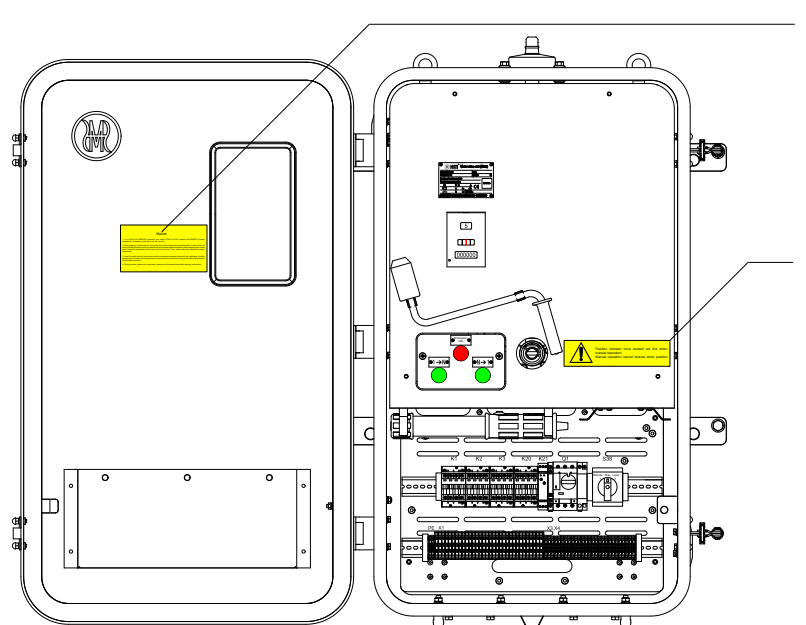
Potential hazards	Description	Instructions
Electric shock	Contact with live electrical parts can cause serious injury or death.	- Before inspection or maintenance, take the following safety actions: 1. Cut off the power supply for the equipment. 2. Prevent the equipment from re-energization. 3. Verify that no voltage is present. 4. Discharge, ground, and short circuit as required. 5. Isolate adjacent live parts to prevent accidental contact. - For the specific task, wear the proper Personal Protective Equipment (PPE), for example, safety helmets and work gloves.
Improper grounding	Improper grounding can cause dangerous potential difference and equipment fault.	Before installation, operation, or testing, ensure correct grounding.

Table 2-1 Potential danger and preventive measures (continued)

Potential hazards	Description	Instructions
Explosion or fire	Highly flammable or explosive gases, vapors, and dust can cause serious explosions and fires.	• Operate the product only in a non-explosive environment. • Keep oil away from open flames, sparks, or any potential fire sources. • In case of a fire, use dry powder, foam, or carbon dioxide extinguishers. Never use water.
Risk of accidental button contact	If not disconnecting the CMA7 power supply during inspection and repair, accidental button contact can cause pinch or collision injury by moving parts.	Always disconnect the power supply before inspection and repair.
Mechanical hazards	Moving or rotating parts can cause entanglement risks.	• Wear well-fitting clothing. • Do not wear rings, necklaces, or other jewelry. • Those having long hair must wear a hairnet.
Work area hazards	Poor lighting and a cluttered area can increase the risk of accidents.	• Keep the work area clean and well-lit. • Obey national accident prevention regulations.
Noise hazards	Tap-change operation can produce high noise levels.	Wear hearing protection as required.
Waste disposal	Incorrect disposal of waste materials, such as packaging or removed parts, can harm the environment and lead to health risks.	• Dispose of waste according to local regulations. • Different waste categories require specific disposal methods. Improper waste disposal can harm the environment and human health.

2.2. Safety labels

The product adopts the following safety labels.



Safety label	Overview of the content
1	- For LOCAL and REMOTE operation, turn switch S38 to LOCAL position and REMOTE position respectively. Otherwise, motor drive unit will not work. - When applying 3-phase power to the motor drive unit and press the operating button, if motor drive unit trips right after several revolutions, please adjust the power phase sequence (interchange any two phase wires), manually operate the motor drive unit to the red line. Then restart and check whether the motor drive works. - Insert the cable into the motor drive unit from the bottom entrance hole and finish installation, stuff the gap between the cable and entrance hole to avoid moisture and dust coming into the cabinet affecting the life of electric elements. - During manual operation by hand crank, please do not operate further after reaching end position.
2	Position indicator must exceed red line when manual operation! Manual operation cannot reverse when position indicator doesn't reach red line!

Figure 2-1 Safety labels on the product

⚠ WARNING



Risk of structural damage or personal injury!

Missing safety labels can prevent recognition of hazards.

- ✓ Keep all safety labels intact and legible.
- ✓ If a label is missing or damaged, replace it immediately.

2.3. Intended use

This product is the CMA7 Motor Drive Unit designed for driving and controlling the tap-change operation of a transformer tap-changer.

It is intended for use in electrical power systems and related installations.

When operated in accordance with the instructions, conditions, and safety warnings provided in this manual and on the product itself, it does not pose a hazard to personnel, equipment, or the environment.

This assurance applies throughout the product's entire lifecycle—from delivery and installation to operation, maintenance, and final disposal.

The following table shows the intended use of this an on-load tap-changer product.

Table 2-2 Intended use

Aspect	Requirements & specifications
Equipment compatibility	Install and operate this product only with the transformer and the tap-changer as specified in the order documentation. If this product and accessories are supplied as a complete set, verify that the serial numbers of all components match and are compatible.
Product identification	Use the nameplate data (compliance standards, model, serial number, year of manufacture) to verify specifications before installation. For the location of this nameplate, see 3.6 Nameplate .
Operating conditions	Operate this product strictly under the environmental and electrical conditions defined in 3.3 Technical specifications and 3.4 Operation conditions . These include ambient temperature, humidity, and voltage range.
Personnel qualification	All installation, commissioning, inspection, maintenance, and disposal work must be performed by personnel qualified as described in 2.5 Personnel qualifications . Operators must ensure: • Regular safety training regarding procedures and guidelines. • Safety instructions and emergency protocols are displayed and accessible in work area.

Table 2-2 Intended use (continued)

Aspect	Requirements & specifications
Operating procedures	Operators must: • Strictly follow all warnings and safety instructions as described in this manual. • Strictly follow procedures for transportation, installation, commissioning, testing, and disposal as described in this manual. • Operate this product exclusively only when in good working condition. Regularly inspect safety devices to ensure reliable operation. • Provide all necessary tools and PPE for each task as described in this manual.
Tools & equipment	Use the provided tools and equipment solely for their intended purpose and in accordance with the instructions in this manual. Do not modify or misuse the tools.
Auxiliary equipment	The use of auxiliary equipment such as an oil purification system is optional and depends on the specific application and customer requirements. Only use auxiliary equipment approved by the manufacturer.
Spare parts	Only use spare parts and consumables that have been approved by the manufacturer to ensure continued safety and functionality.
Modifications	Any modification or alteration of the product that has not been authorized by the manufacturer is strictly prohibited. Unauthorized modifications may lead to personal injury, equipment failure, or loss of warranty.

2.4. Foreseeable misuse

Any use of this product beyond the scope of intended use as described in [2.3 Intended use](#) is considered improper and may pose significant hazards to personnel, equipment, and the environment.

Foreseeable forms of misuse include, but are not limited to:

- Operation with incompatible equipment: Using this product with transformers, Tap-changers, or accessories that have not been specified in the order documentation.
- Use outside specified environmental conditions: Operating this product in environments that exceed the permitted temperature, humidity, or other conditions specified in [3.4 Operation conditions](#).
- Unauthorized modifications: Making mechanical or electrical modifications to this product or its components without explicit written approval from the manufacturer.
- Bypassing safety devices: Disabling or tampering with protective relays, or warning systems integrated into this product or associated equipment.
- Improper electrical connections: Incorrect wiring or voltage application that does not comply with the connection diagrams or rated electrical values.
- Neglecting maintenance and inspection intervals: Failing to carry out the prescribed inspections and maintenance tasks, which could lead to unsafe operating conditions or premature failure.

- Inadequate qualification of personnel: Allowing untrained or unauthorized individuals to install, commission, maintain, or operate this product.
- Stepping on or placing weight on critical components: Ignoring safety labels may result in mechanical damage or personal injury.
- Unauthorized or inappropriate changes to this product may lead to personal injury, material damage and operational faults.

2.5. Personnel qualifications

All personnel involved in the installation, commissioning, operation, and maintenance of the product must meet the qualification levels in the table below.

Table 2-3 Personnel qualifications

Qualification level	Applicable positions	Core competencies and requirements
Electrical Technical Personnel	• Technical Field Assistant (TFA) • Commissioning Engineer • Maintenance Technician	• Holding valid technical certifications as required by national safety regulations (e.g., Certified Electrician/Registered Engineer) • Possessing specialized knowledge in identifying and avoiding electrical hazards (e.g., electric arc and high-voltage risks) • Demonstrating proficiency in electrical system operation and maintenance • Understanding local electrical codes and accident prevention regulations • Receiving complete training for operation in specific environments • (TFA and Commissioning Engineers) must complete internal training on transformer installation.
Trained Electrical Personnel	Operator (working under supervision)	• Receiving task-specific training from electrical technical personnel, covering: ◦ Equipment-specific hazards ◦ Safety procedures and protective measures ◦ High-voltage equipment operation • They must perform work under the direct supervision of electrical technical personnel
Maintenance Technician	Maintenance Technician	Holding certification in Mechanical and Electrical Safety Regulations
All positions	All personnel	Possessing fundamental knowledge of power transformers and on-load tap-changer systems

2.6. Personal Protective Equipment (PPE)

The following table lists the mandatory personal protective equipment during operation and maintenance to mitigate potential risks.

Table 2-4 Personal Protective Equipment

	Wear close-fitting workwear made of low-tear-strength material, with tight cuffs and no protruding parts, to minimize the risk of being caught by moving mechanical components.
	Wear safety shoes with puncture-resistant and slip-resistant soles.
	Use protective goggles to shield the eyes and the face from impact, dust, splashes, and high-temperature hazards.
	Wear a safety helmet to protect the head against impact.
	Use hearing protection in noisy environments to prevent hearing impairment. Certain operations of on-load tap-changers, especially during switching sequences, functional tests or fault conditions, may generate impulse noise levels exceeding 85 dB(A). Repeated or prolonged exposure to such noise levels can cause permanent hearing impairment.
	Wear insulated gloves to protect your hands from mechanical, thermal and electrical hazards.

3. Product Overview

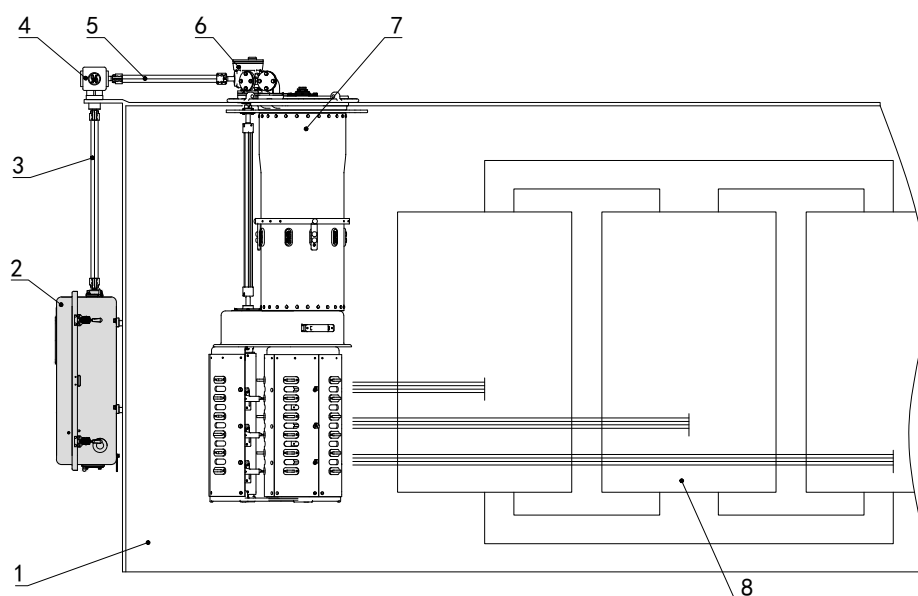
CMA7 Motor Drive UnitIt is a drive and control unit specifically designed for the tap-change operation of a tap-changer.

This unit integrates all electrical and mechanical components required to drive the tap-changer within a single housing. It adopts the step-by-step control principle: Each tap-change operation is initiated by a single control signal and automatically stops after the operation is completed.

CMA7 Motor Drive UnitIt is typically Installed on the wall of the transformer tank and connected to the tap-changer via components of the drive mechanism. This ensures reliable power transmission and position control.

3.1. Position of the CMA7 in the transformer

The CMA7 is typically installed the wall of the transformer tank and connected to the tap-changer via components of the drive mechanism. This enables electric and remote tap-change operations.



1. Transformer tank	2. CMA7 Motor Drive Unit (MDU)
3. Vertical driving shaft	4. Bevel gearbox
5. Horizontal driving shaft	6. Top gearbox
7. Tap-changer main body	8. Transformer winding

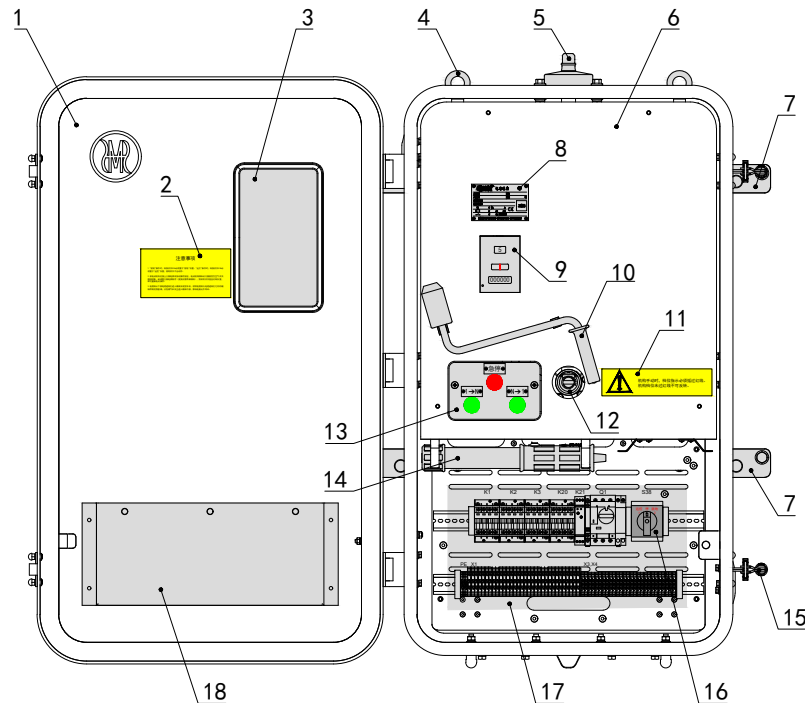
Figure 3-1 Position of the CMA7 in the transformer

3.2. Composition and structure of the CMA7

CMA7 Motor Drive UnitIt is mainly composed of the following parts:

- Housing (tank door and tank)
- Drive mechanism
- Position indication mechanism
- Electrical control components

The figure below shows the composition and structure of the CMA7.



1. Enclosure door	2. Caution label	3. Viewing window for indication area
4. Lifting lug	5. Output shaft	6. Drive mechanism cover
7. Mounting plate	8. Nameplate	9. Indication area
10. Hand crank	11. Safety label	12. Hand-crank access port with interlock switch
13. Local control switch (including emergency stop, 1→N (S1) and N→1 (S2))	14. Portable lamp	15. Door latch
16. Control selector switch (S38)	17. Electrical control components	18. Document pocket

Figure 3-2 Composition and structure of the CMA7

3.3. Technical specifications

No.	Item		Data		
1	Motor power (kW)		0.75	1.1	2.2
2	Rated voltage (V) ¹	Three-phase	380	380	380
		Single-phase	220	220	220
3	Rated current (A)	Three-phase	2.0	2.8	5.1
		Single-phase	3.4	5	8.8
4	Rated frequency (Hz)		50, 60	50, 60	50, 60
5	Synchronous speed (rpm)		1,400	1,400	1,400
6	Number of driving shaft revolutions per tap-change operation (circle)		33		
7	Motor operation time per tap-change operation (circle)		About 5 s		
8	Torque of driving shaft (N·m)		18	26	52
9	Maximum operation position number ²		35		
10	Control and heater voltage (V) ¹		220		
11	Power consumption of control circuit (W)	Start-up	52		
		Operation	24		
12	Heater power (W)		50		
13	Voltage test to ground (kV)		2/(1 min)		
14	Weight (kg)		90		
15	Mechanical lifespan		More than 800,000 operations		


Note:

1. The rated voltage and the control and heater voltage can be customized according to customer requirements. For the actual product voltage value, refer to the order requirements.
2. The number of operation position should correspond to that of the on-load tap-changer.

3.4. Operation conditions

CMA7 Motor Drive Unit Be sure to operate under the following conditions:

Operation conditions	Requirements
Range of ambient temperature	-45°C to +55°C
Ground verticality deviation of the CMA7 after installation on the transformer	≤ 5%
Requirements for installation locations	No excessive dust, explosive gases or corrosive gases
Altitude	≤ 2,000 m



Note:

If customized solutions are required for special environment conditions, contact Huaming.

3.5. Requirements for product compatibility

To ensure safe and reliable operation, CMA7 must correctly match key system components. Compatibility with the transformer, tap-changer, and mounting interface is critical for achieving optimal performance and avoiding damage or potential safety hazards.

- Tap-changer: Huaming tap-changer



Important:

For detailed information on compatible component models, refer to the corresponding order document. When using non-Huaming components, secondary development may be required.



3.6. Nameplate

CMA7 Motor Drive UnitIts nameplate is on the tank cover. The nameplate provides key identification information, including the serial number, date of manufacture, model, technical specifications, and applicable compliance standards.

Motor drive unit (MDU)

Product number

Date

Product model

Weight **kg**

Number of equipped switch

Model of equipped switch

U_i

V

I_u

A

V

Hz

kW

V

Hz

Number of circuit diagram

Executive standard: GB 10230.1-2019 IEC 60214-1:2014 IEC 61439-2:2020

QR code

Figure 3-3 CMA7 nameplate

4. Packaging, Transportation, and Storage

4.1. Supplied components

The CMA7 Motor Drive Unit (MDU) is delivered as a complete set, including the following components:

1. CMA7 Motor Drive Unit (MDU)
2. Controller (optional)
3. Accessories for customers' special requirements (if any)



Note:

The actual supplied components are based on the purchase order. For the specific delivery condition, refer to the packing list.

4.2. Transportation

WARNING



Risk of Toppling!

A falling or toppling package may cause severe personal injury and equipment damage.

- ✓ Only trained and designated personnel may select slings and secure loads.
- ✓ Do not stand under a suspended load.
- ✓ Use transport and lifting equipment with a load capacity greater than 1,000 kg.

CAUTION



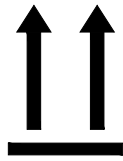
Risk of improper transport!

Improper handling may lead to product damage, functional failure, or secondary safety incidents.

- ✓ Transport and lifting must be carried out by qualified personnel.
- ✓ Only use the lifting and transport equipment with a rated load capacity not below 1,000 kg.
- ✓ Only use the designated lifting points marked on the package.
- ✓ Secure the packaging case firmly during the transport and lifting process. Prevent vibration, impact, tilting, falling, or collision to avoid product damage.
- ✓ If the package has dropped or sustained a severe impact, check whether CMA7 is damaged. If necessary, contact Huaming for inspection and repair.

The factory packaging is suitable for multiple transport methods. Always follow these instructions during transport:

- Always position the package according to the mark of the center of gravity.
- During storage, stack packages only when the load on the top surface does not exceed 500 kg/m².
- Always follow the markings on the package during transport (as shown in the figure below).



This side up

Gross weight (G.W.)

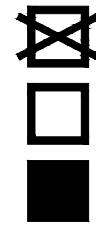


Handle with care

Net weight (N.W.)



Protect from rain
and moisture



Stacking layer limit

Kilogram (KG)

4.2.1. Receiving inspection

Upon receiving the packages, inspect each package and verify that the packaging is intact and undamaged. If any damage is found (for example dents, tears, moisture), perform these steps:

1. Refuse delivery and immediately notify the carrier.
2. Take photos of the damaged packaging and contents if possible.
3. If the issue cannot be resolved with the carrier, contact Huaming After-Sales Service in writing.

4.2.2. Transshipment

After the receiving inspection, you may need to transship the packages to another location.

1. Unload and handle the packaged unit in accordance with the instructions on the packaging.
2. Use the designated lifting points for hoisting or follow the forklift markings when transporting with a forklift.



Important:

Select an appropriate forklift and/or crane based on the dimensions and gross weight of the crate or package.

4.2.3. Unpacking and inspection

To unpack and inspect the packages, perform these steps:

1. Before unpacking, move the package to the installation site.



Important:

Take care not to damage the original packaging during unpacking and inspection.

2. Inspect each package against the packing list and Confirmation of Acceptance form. Verify the following:

- The quantity of items matches the packing list.
- The product specifications match the order documentation.

! Important:

- Missing components may affect subsequent operation. If any item or accessory does not match the packing list, contact Huaming immediately.
- If the package has been dropped or subjected to impact after delivery, assume potential damage. Thoroughly inspect the CMA7. If necessary, contact Huaming to arrange inspection and/or repair by qualified personnel.

3. Before installation, store all components in a dry location and keep CMA7 sealed in its original packaging.

4.2.4. Lifting

 WARNING	
	The lifting process involve risks of personal injury and equipment damage! Improper lifting may lead to load falling or instability, causing serious personal injury or equipment damage. ✓ Before lifting, ensure the lifting device is functioning properly, the CMA7 lifting lugs are intact, and the enclosure doors are closed and locked. ✓ Use the lifting equipment with a rated load capacity not below 1,000 kg. ✓ Avoid sudden movement or shaking when lifting or lowering the load. ✓ Do not stand or walk under a suspended load.

Use the lifting lugs on the top of the CMA7 cubicle to lift the CMA7 vertically.

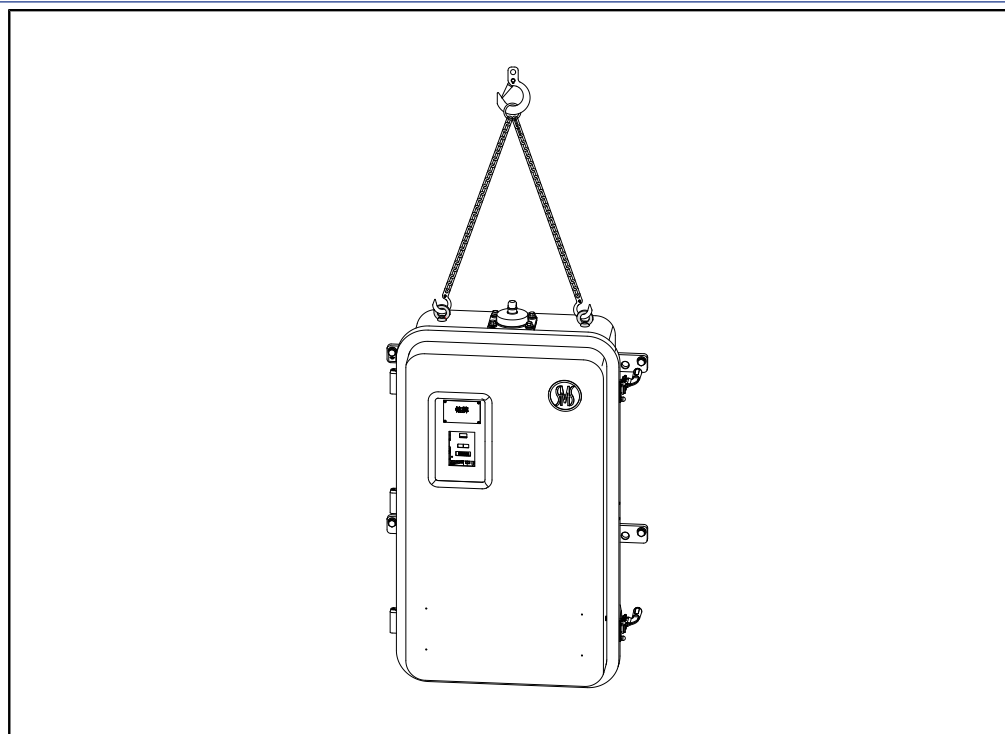


Figure 4-1 Lift the CMA7

4.3. Storage

Specially-sealed units can be stored outdoors if the following conditions are met:

- The ambient temperature is between -25°C and +40°C.
- The storage area is away from flammable, explosive, or corrosive gases.
- Protect the CMA7 against moisture, dust, rodents, and insects.
- Provide the corresponding protection, to protect the CMA7 against moisture, dust accumulation, rodents and insects.
- Regularly check the stored equipment for any abnormalities.
- Regularly replace desiccants during long-term storage and ensure the packaging remains sealed.

5. CMA7 Installation on the Transformer

5.1. Install CMA7

Prepare installation tools and four fastening bolts.

The CMA7 has been assembled with its mounting plate when delivery. Install bolts through the holes in the mounting plate to fix the CMA7 on the transformer or another location.

NOTICE

Incorrect CMA7 installation!

Incorrect CMA7 installation will affect tap-changer operation.

- ✓ Connect the CMA7 to the tap-changer only when they are in the set position. Refer to the tap-changer wiring diagram for the set position.
- ✓ Ensure that the CMA7 mounting plate is flat to prevent it from deformation. Otherwise, it will affect operation.

⚠ WARNING



The lifting process involve risks of personal injury and equipment damage!

Improper lifting may lead to load falling or instability, causing serious personal injury or equipment damage.

- ✓ Before lifting, ensure the lifting device is functioning properly, the CMA7 lifting lugs are intact, and the enclosure doors are closed and locked.
- ✓ Use the lifting equipment with a rated load capacity not below 1,000 kg.
- ✓ Avoid sudden movement or shaking when lifting or lowering the load.
- ✓ Do not stand or walk under a suspended load.

1. Use the lifting lugs on the top of the CMA7 cubicle to lift the CMA7 vertically.

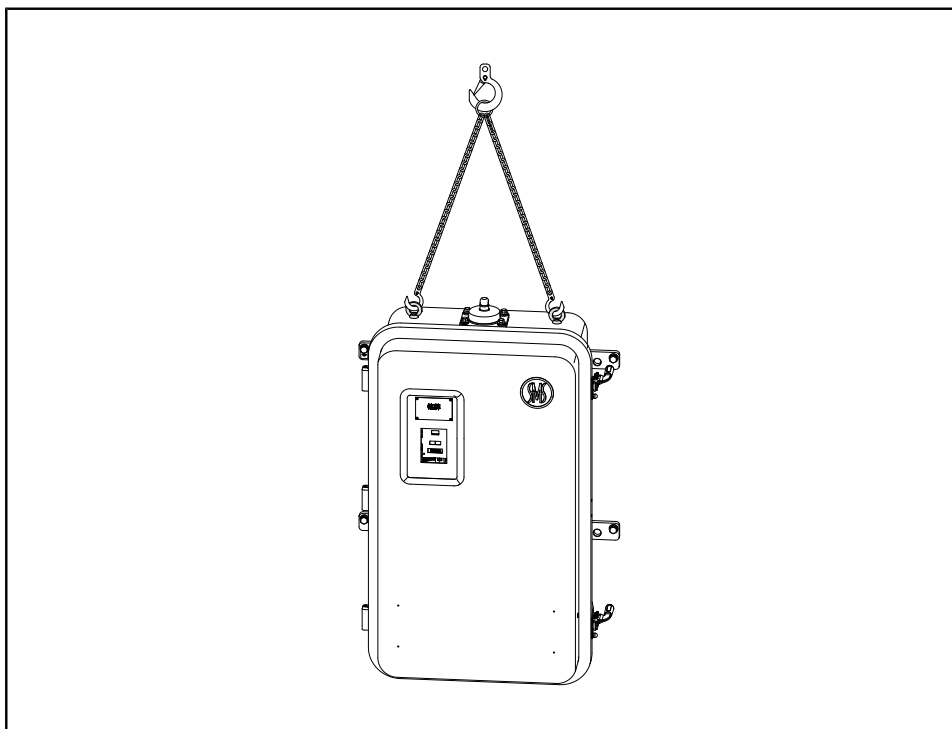


Figure 5-1 Lift the CMA7

2. Use four fastening bolts to fix the CMA7 vertically on the transformer tank wall.

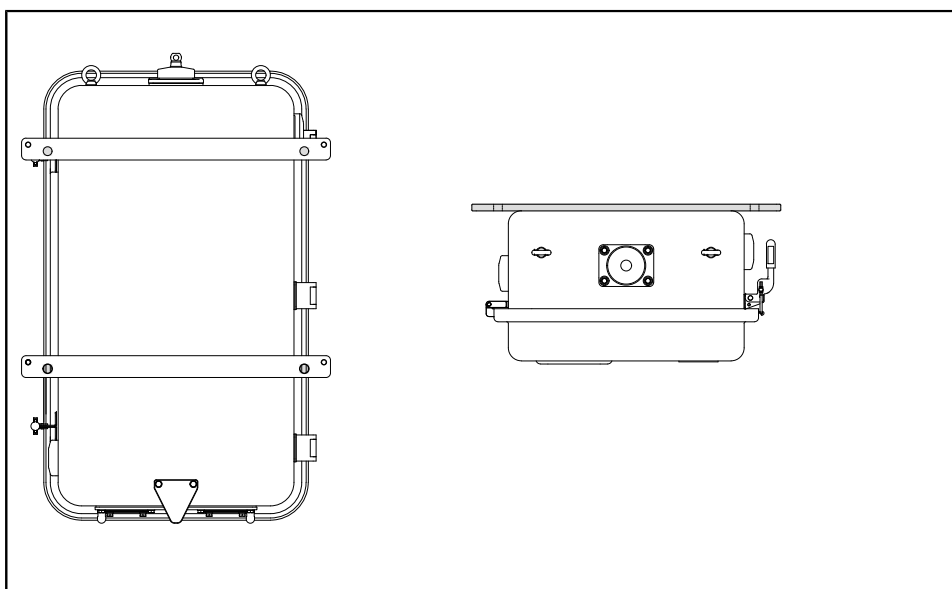


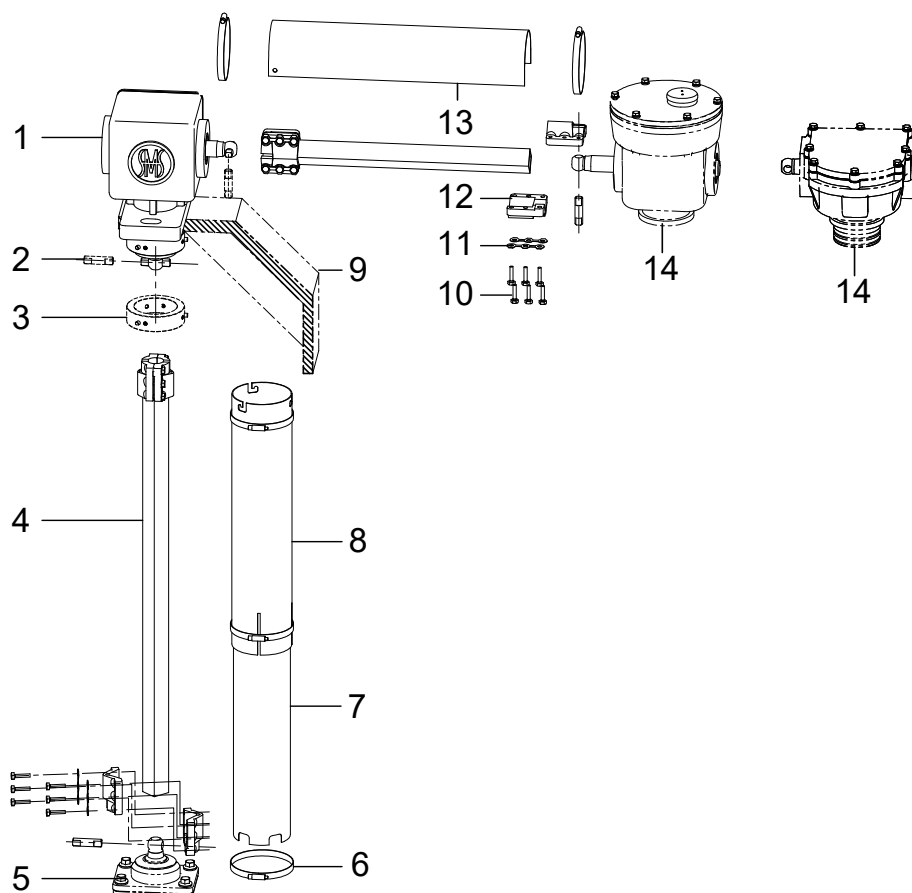
Figure 5-2 Vertically fix the CMA7

3. **Optional:** If the transformer has strong mechanical vibration, it is recommended to install vibration dampers between connecting holes and the transformer tank wall.

5.2. Install drive mechanism components

5.2.1. Drive mechanism components

Drive mechanism components consist of the bevel gearbox, driving shafts, couplings and other accessories.



1. Bevel gearbox	2. Cylindrical pin	3. Bushing
4. Driving shaft	5. Motor Drive Unit (MDU)	6. Clamp
7. Vertical guard 1	8. Vertical guard 2	9. Gearbox bracket (transformer)
10. M6 screw	11. Lock washer	12. Coupling
13. Horizontal guard	14. Top gearbox	

Figure 5-3 Drive mechanism components

5.2.2. Adjust the top gearbox

NOTICE

Mismatched OLTC components!

Mismatched components can affect the normal and stable operation of the OLTC.

✓ Ensure that the nameplate numbers match. Use each gearbox only with its designated tap-changer.



Important:

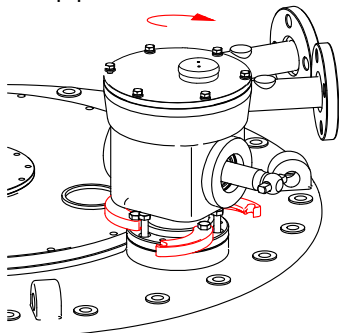
For a driving shaft longer than 2 m, install an intermediate support gearbox.

To adjust the top gearbox, perform these steps:

1. Loosen the six bolts securing the clamping plate beneath the top gearbox. (Do not remove the bolts completely.)
2. Pry the three clamping plates outwards, and then rotate the top gearbox to align its output shaft with the CMA7.

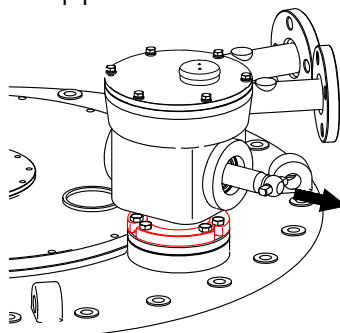
3. Reposition the clamping plate and tighten the six bolts, ensuring that the top gearbox is securely mounted without any looseness. Tightening torque: 15 Nm.

The state when loosening all clamp plate bolts:

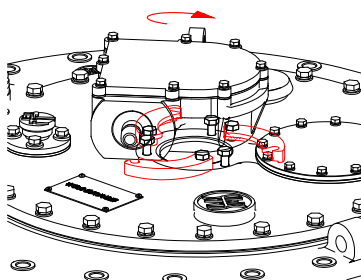


The top gearbox can rotate in a circle in the direction of the arrow

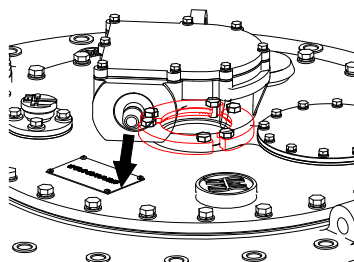
The state when tightening all clamp plate bolts:



The horizontal output shaft points toward the MDU



The top gearbox can rotate in a circle in the direction of the arrow



The horizontal output shaft points toward the MDU

Figure 5-4 Adjust and secure the top gearbox

5.2.3. Install the bevel gearbox

NOTICE

Mismatched OLTC components!

Mismatched components can affect the normal and stable operation of the OLTC.

✓ Ensure that the nameplate numbers match. Each gearbox corresponds to a specific OLTC.



Important:

For driving shafts longer than 2 meters, install an intermediate support gearbox.

1. Align the shaft of the bevel gearbox vertically with the shaft of the CMA7. Make sure that they are precisely aligned.

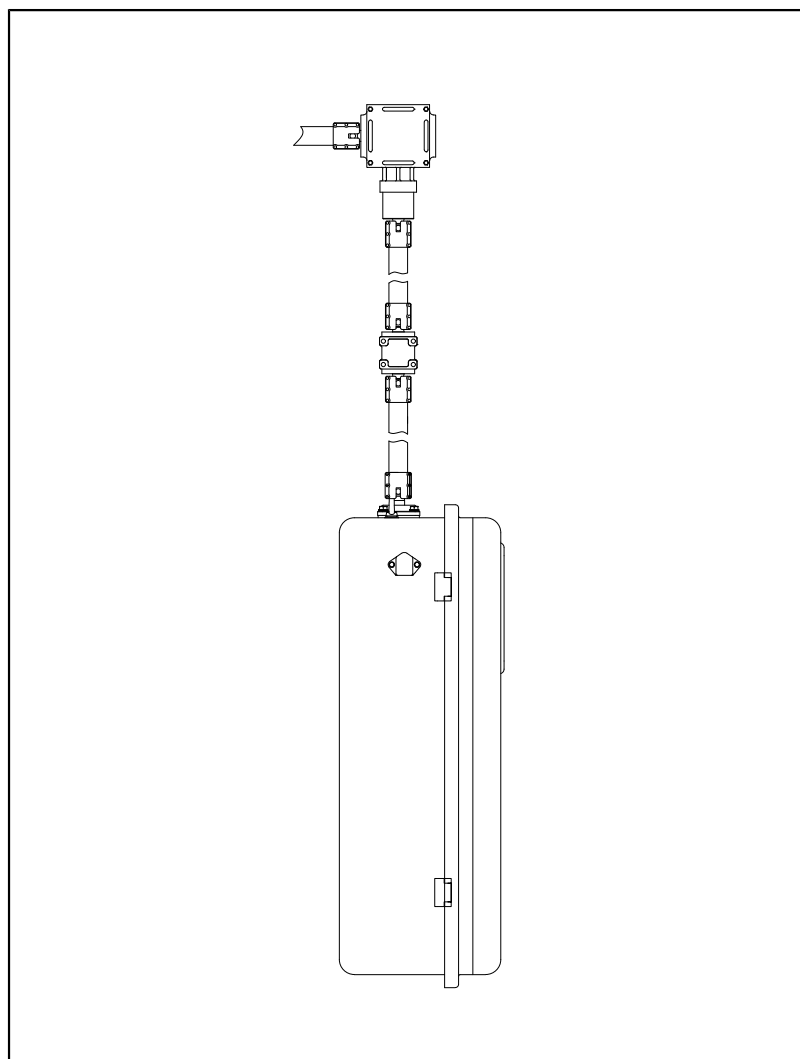


Figure 5-5 Align the shaft of the bevel gearbox and the shaft of the CMA7

2. Secure the bevel gearbox on the transformer cover bracket with two bolts (18 mm diameter).

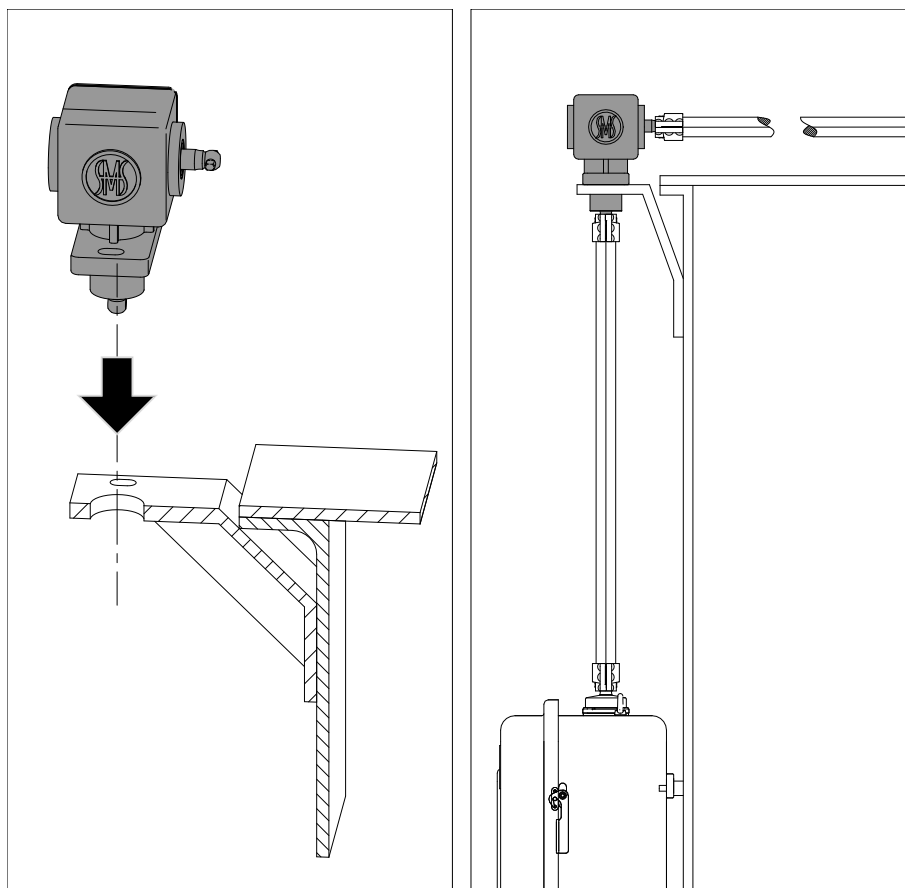


Figure 5-6 Install the bevel gearbox

5.2.4. Install the horizontal driving shaft and vertical driving shaft



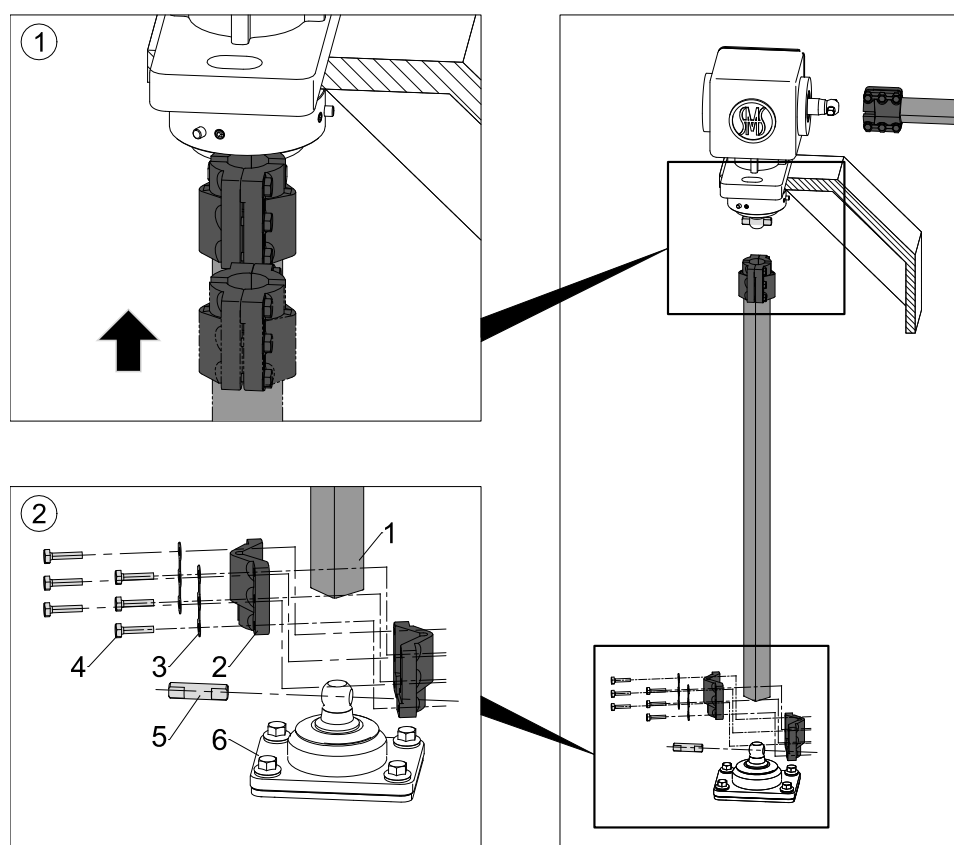
Important:

- For a driving shaft longer than 2 m, install an intermediate support gearbox.
- To install the vertical guard, install it when installing the vertical driving shaft. See [5.2.5.0 Install the vertical guard](#) for details.

Connect the driving shaft, the CMA7 and the tap-changer.

The figure below shows the connection of the coupling between the bevel gearbox and the vertical driving shaft. The installation modes of the horizontal driving shaft coupling and vertical driving shaft coupling are the same.

Install six M6 screws using a 10 mm wrench or socket, tightening torque: 9 Nm.



1. Driving shaft	2. Coupling	3. Lock washer
4. M6 screw	5. Cylindrical pin	6. CMA7

Figure 5-7 Install the bevel gearbox and the driving shaft

5.2.5. Install the driving shaft guards



Important:

For a driving shaft longer than 2 meters, install an intermediate support gearbox.

Install the horizontal guard

The length of the horizontal guard is determined by the transformer manufacturer.

1. After installing the driving shaft, place the associated clamps on both ends of the horizontal guard.
2. Position the guard on the mounting surfaces of the top gearbox and the bevel gearbox. Slide the clamps onto the mounting surfaces and tighten them.

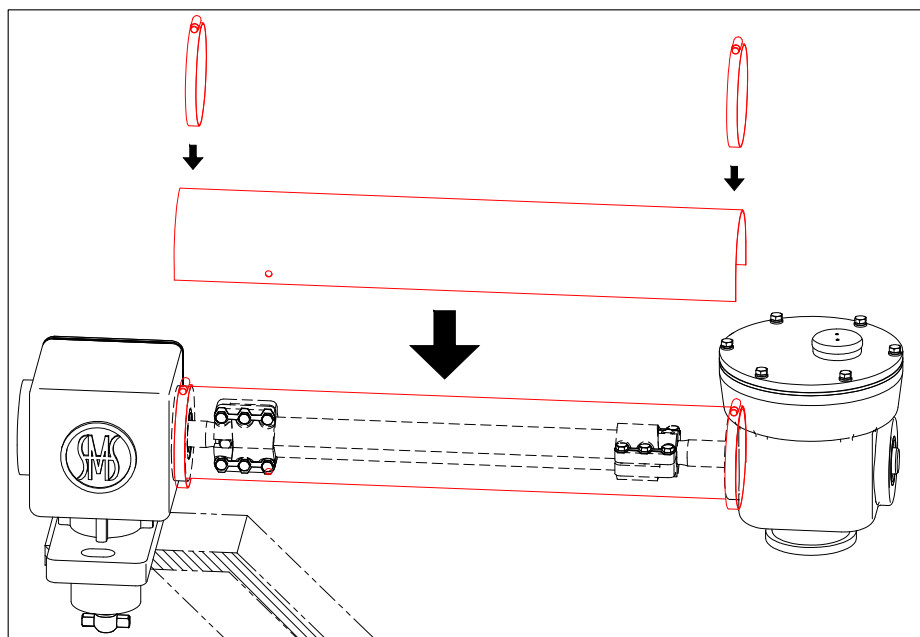


Figure 5-8 Install the horizontal guard

Install the vertical guard

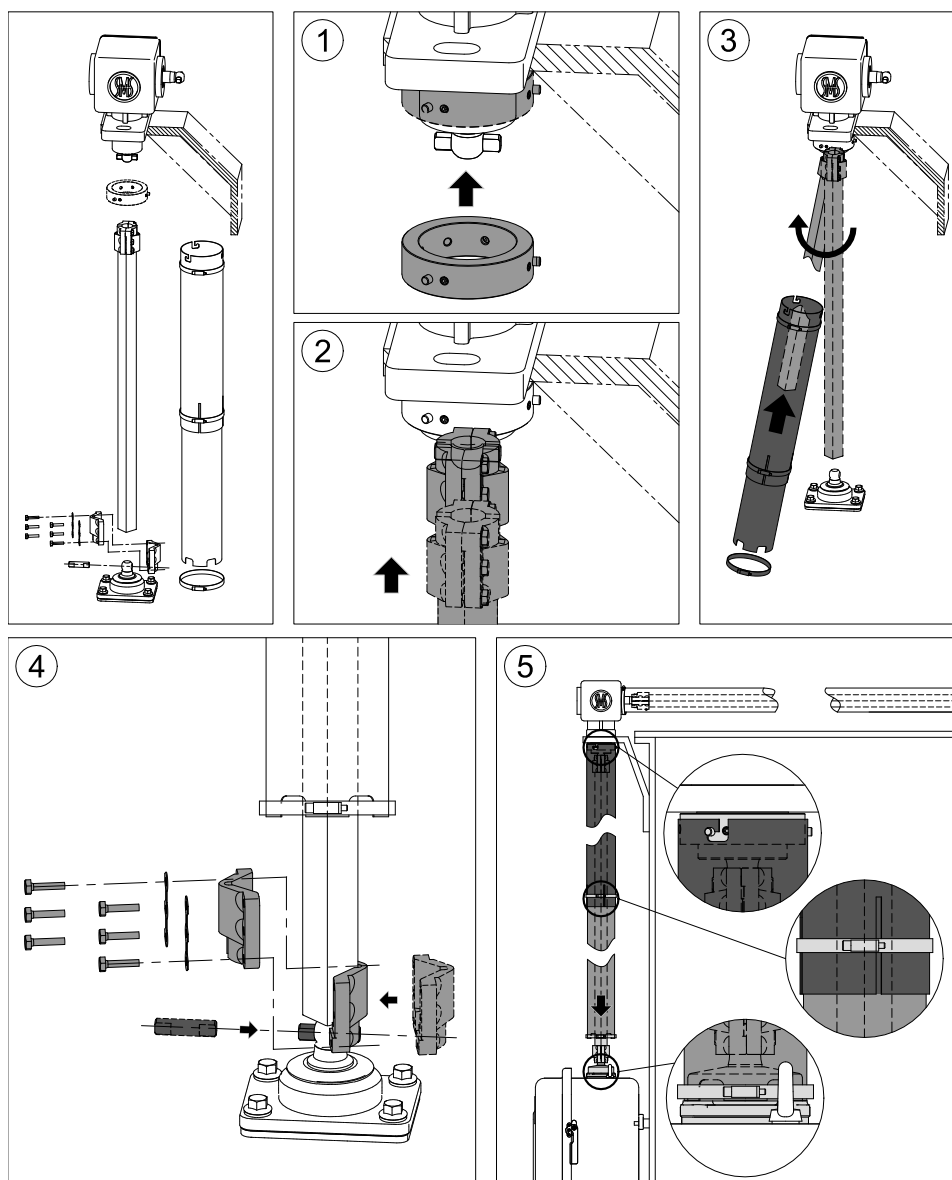


Figure 5-9 Installation steps for the vertical guard

To install the vertical guard, perform these steps:

1. Install the three 6 x 16 stainless steel cylindrical pins into the bushing (evenly spaced at 120° intervals). The pins should protrude approximately 5 mm beyond the outer diameter of the bushing and should not protrude into the inner diameter.
2. Position the bushing at the lower end of the bevel gearbox. Secure it with four M6 x 10 locating screws (evenly spaced at 90°). Tightening torque: 9 Nm.

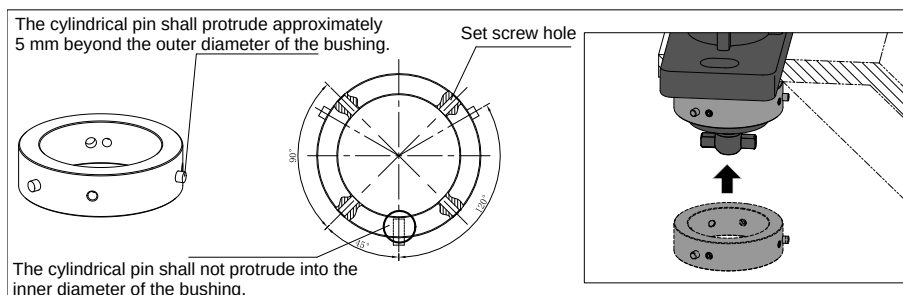


Figure 5-10 Install the bushing

3. Slide the vertical guard upward onto the lower end of the bevel gearbox, ensuring it sleeves over the bushing. Align the three grooves on the guard with the three cylindrical pins. Rotate the guard until the pins snap into the grooves (see [Figure 5-11](#)).
4. Slide the thinner lower guard onto the output shaft of the CMA7 (see [Figure 5-11](#)).

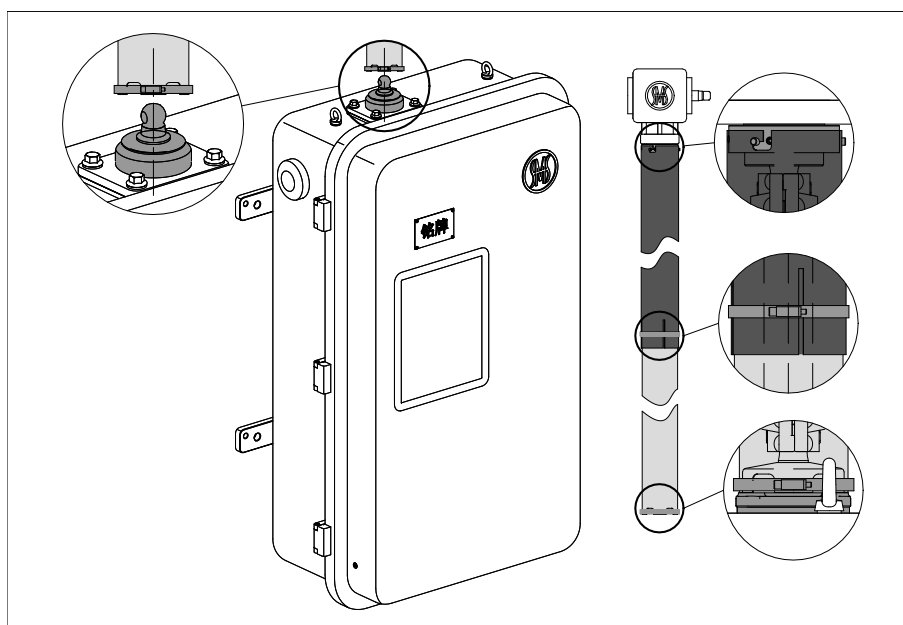


Figure 5-11 Install the vertical guard

5. After positioning both guards in place, secure them with two clamps (see [Figure 5-11](#)).
6. If the vertical driving shaft exceeds 2 m and an intermediate connecting shaft is used, repeat Step 1 to Step 5.

5.3. Calibrate the drive system of the tap-changer

Before electrically operating the CMA7, manually operate the tap-changer through one complete cycle. This can verify whether it is aligned and functions correctly. Ensure the time interval between the diverter switch operation and the completion of the CMA7 movement is the same in both rotation directions.

To ensure reliable operation of the tap-changer, calibrate the drive system after disconnecting and reconnecting the horizontal or vertical shaft.

5.3.1. Calibration procedure

Calibrate the drive system in the following steps:

1. Use the hand crank to rotate the driving shaft in the 1→N direction. When the diverter switch operation is heard ("click" sound), continue to rotate and count the number of hand crank rotations (m). Until: The green field appears in the CMA7 tap position indicator area; The red marker is centered in the window.
2. Rotate the hand crank in the opposite direction (N→1) back to the original set position. Use the same method to count the number of hand crank rotations (k).
3. If $m = k$, the connection is correct. If $m \neq k$ and the absolute difference $(|m - k|) > 1$: Disconnect the vertical driving shaft; Rotate the hand crank $1/2 (|m - k|)$ rotations toward the direction with more rotations. Then, reconnect the vertical driving shaft to the CMA7.
4. Repeat the steps above, until the number of rotations is nearly equal in both directions.
5. After calibration, rotate the red marker in the tap position indicator area to the center of the window.

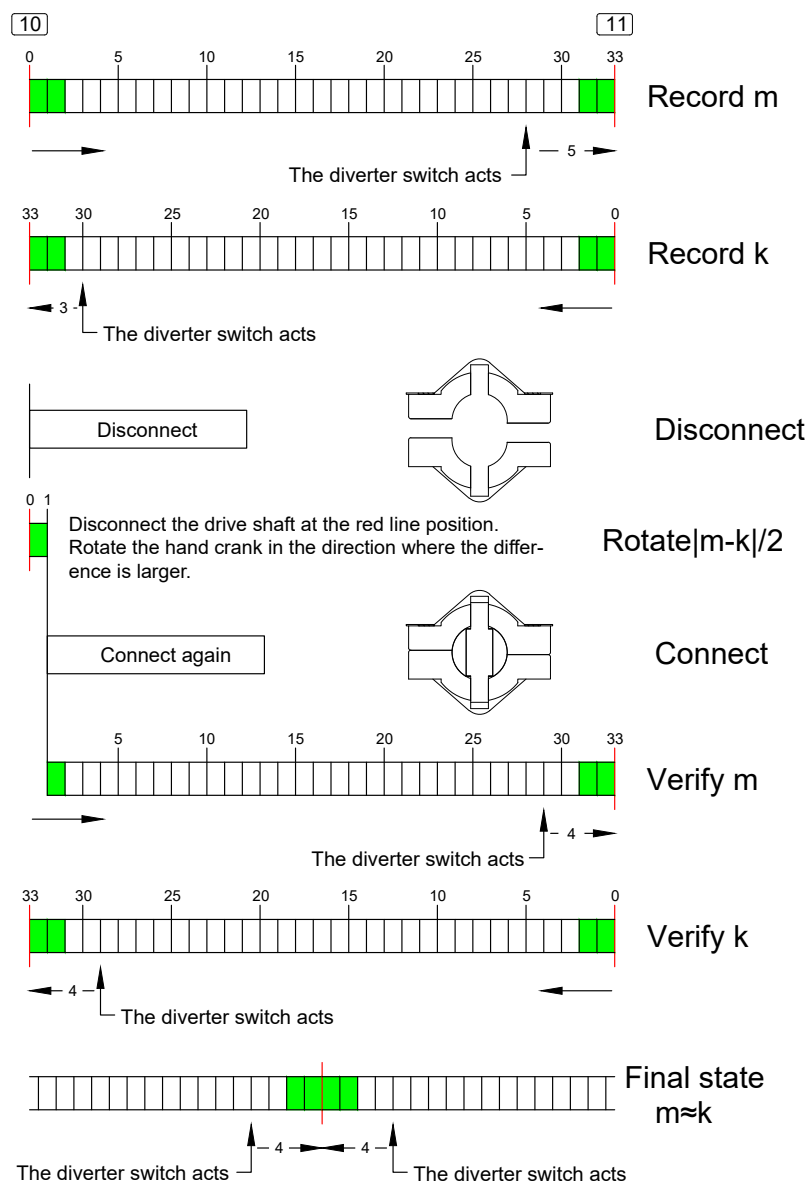


Figure 5-12 Calibration schematic diagram

5.3.2. Calibration example

To calibrate a tap-changer with the 10193W tap selector:

1. Rotate from Position 10 (set position) to Position 11 ($m = 5$ rotations).
2. Rotate from Position 11 back to Position 10 ($k = 3$ rotations).
3. Calculate:
 - a. Difference in the number of rotations: $|m - k| = |5 - 3| = 2$ rotations
 - b. Adjustment of the number of rotations: $1/2 (m - k) = 1/2 (5 - 3) = 1$ rotation
4. Disconnect the vertical driving shaft from the CMA7. Rotate the hand crank one full rotation from Position 10 to Position 11. Then, reconnect the shaft.

5. Verify whether the difference in the number of rotations is balanced in both directions.
6. After calibration, rotate the red marker in the tap position indicator area to the center of the window.

Calibration summary:

1. Record the number of rotations in both directions (m and k).
2. Disconnect the coupling, and rotate the hand crank $1/2 (|m - k|)$ rotations toward the direction with more rotations.
3. Reconnect and verify until $|m - k| < 1$.
4. Rotate the red marker in the tap position indicator area to the center of the window.

6. Transformer Transportation and On-Site Commissioning

6.1. Remove the CMA7 for transportation

When transporting the transformer from the manufacturer to the site, if the CMA7 must be removed because the overall dimensions exceed transportation limits, store and secure the tap changer and the CMA7 in their as-delivered condition.



Important:

Do not operate the CMA7 when it is disconnected from the tap-changer.

To remove the CMA7 for transportation, perform these steps:

1. Set the CMA7 to its set position.
2. Disconnect the coupling.
3. Remove the CMA7.

After transporting the CMA7 to the site, reinstall the CMA7 and components of the drive mechanism.

6.2. CMA7 functional test

Before putting the CMA7 into operation, conduct a functional test. This test will confirm that the control and drive systems operate correctly according to the design requirements. The functional test aims to make sure that the CMA7 has the tap-change ability and protection action. Do this test before using the CMA7 at the site. This will prevent damage to the equipment from faults in the control circuit, motor operation, or limit switches.

The functional test mainly includes:

- Check whether the step-by-step control function is correct
- Make sure that the mechanical limit device can operate reliably
- Verify whether the electrical limit switch can work

Generally, conduct this test after factory routine tests, installation on site, but before energization. Conduct this test also after overhaul or maintenance as a confirmation step. The functional test will help you find and eliminate potential problems before transformer operation. This will keep the tap-changer and the CMA7 reliable in operation.



Important:

Observe through the viewing window for indication area on the CMA7 enclosure door. Make sure that the position on the CMA7 is the same as the position on the tap-changer indicator.

Check the step-by-step control function

Before powering on the motor circuit, control circuit, and auxiliary circuit, make sure that the voltage, current and value meet the requirements:

The CMA7 adopts a step-by-step control principle:

- Each time pushing the control button, the tap-changer will complete one-step operation automatically. This operation is irrevocable in the halfway (except with the emergency stop).
- Only after the control system finishes the operation and return to the stop position, can you conduct the next tap-change operation.
- You can see the stop position through the viewing window for indication area: When the red line enters the viewing window, it indicates that it is in the stop position.

Check the step-by-step control function according to the operation below:

1. Make sure that **S38** is in the "Local" position.

S38

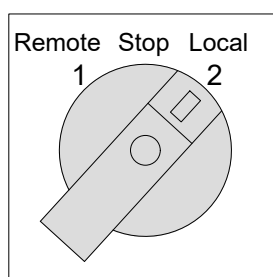


Figure 6-1 S38 in the "Local" position

2. Press down and hold the **1→N** (S1) or **N→1** (S2) button.

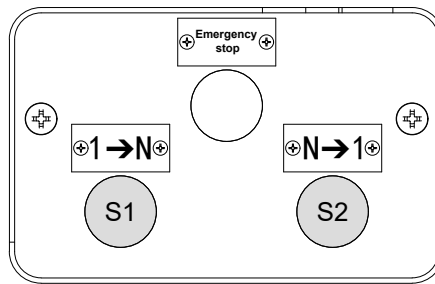


Figure 6-2 1→N (S1) and N→1 (S2) buttons

The CMA7 should only complete one tap-change operation. After the operation, the motor should be able to stop automatically.

Test for mechanical end stop device (mechanical limit)

Before the test for mechanical end stop device, confirm that the CMA7 is in the operation state, namely:

- The CMA7 is in correct connection with the tap-changer. The positions on the CMA7 are the same as the positions on the tap-changer.
- The transformer has no electrical power. You can switch the tap-changer to any position.
- When the drive shaft rotates during the tap-change operation, it will not cause danger to any personnel near it.

Before using the hand crank to operate the CMA7, confirm that the CMA7 meets the manual operation conditions, including:

- The power supply for the CMA7 is off.

1. Make sure that **S38** is in the "Local" position.

S38

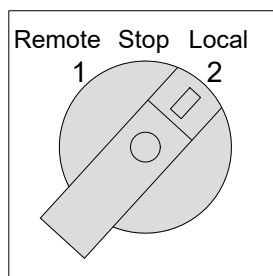


Figure 6-3 S38 in the "Local" position

2. Through electrical operation with the button or manual operation with the hand crank, adjust the CMA7 to the highest or lowest position.
Now, the tap-changer can reach its final working position. But, it must not go beyond the limit position.
3. Use the hand crank and continue to operate it to the limit position.
After turning the hand crank for about 4 to 6 circles, the mechanical limit should work.
4. Turn the hand crank in the opposite direction to release the mechanical limit.
Continue to turn the hand crank to the red line. The CMA7 should return to the last valid position.
5. Repeat the said steps for the other limit position.

The mechanical stop function should operate normally, to effectively prevent the tap-changer from moving past the limit position.

Test for limit switch (electrical limit)

Make sure that the CMA7 and electrical control circuit have correct wiring and meet the conditions for adjustment operation.

1. Move the tap-changer to one limit position.
2. Try to continue the electrical control operation in the same direction. At the highest position, continue to raise the tap position with the raise button **S1**. At the lowest position, continue to lower the tap position with the lower button.
The motor should no longer operate.
3. Try to operate in the opposite direction.
The motor must drive normally.
4. Move the tap-changer to the other limit position. Repeat the said steps.

The electrical limit function must operate normally. It must prevent the motor from operating too much in the same direction.

7. CMA7 Operation

7.1. Remote operation of the CMA7

During normal operation, control tap-change operation through the remote voltage regulator.

1. Confirm the power supply matches the specifications in the corresponding project drawings. Close the motor protective switch Q1, and set the control selector switch (**S38**) to the "Remote" position.
2. Issue one control pulse (raise or lower) from the control room or controller.

The CMA7 automatically completes one full tap-change operation.



Important:

- Even if one more control pulse is issued during the operation, the CMA7 will finish the current operation first.
- If power outage occurs during the operation, the CMA7 will automatically complete the interrupted change after power restoration.

Wait for tap-change to complete before the next operation.

7.2. Local operation of the CMA7

Maintenance or special operation requires direct local operation of the CMA7.



Important:

- Use local operation in special circumstances only, such as commissioning or maintenance.
- Local operation and remote control are interlocked to prevent simultaneous operation.

1. Confirm the power supply matches the specifications in the corresponding project drawings. Close the motor protective switch Q1.
2. Switch the control selector switch (**S38**) to the "Local" position.
3. Use the local control switch **1→N (S1)** or **N→1 (S2)** button to operate the CMA7.

Wait for tap-change to complete before the next operation.

7.3. Use the hand crank to operate the CMA7

If the power supply fails or the CMA7 is abnormal, complete tap-change manually.

1. Remove the hand crank. Insert the hand crank into the hand-crank access port with interlock switch.
The system will automatically disconnect the motor power supply and control power supply.
2. Turn the hand crank clockwise or counterclockwise to operate the tap-changer step by step. (Turn it clockwise to lower the tap position. Turn it counterclockwise to raise the tap position.) While turning the hand crank, listen for the mechanical click sound. Watch the indicator disk (whether the operation indicator passes the red line). Confirm correct engagement of each new position.

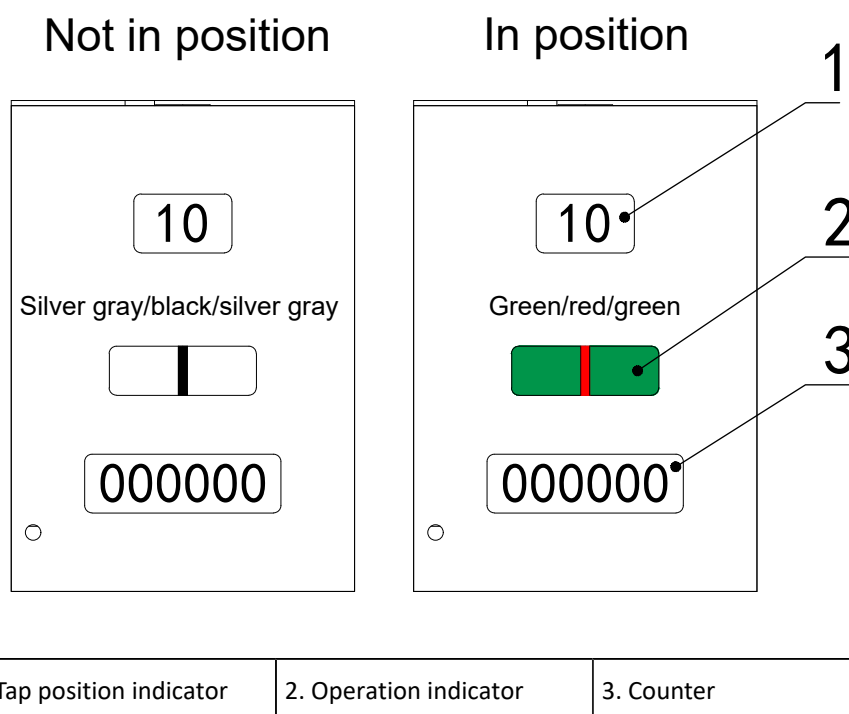


Figure 7-1 The operation indicator passes the red line

The tap position indicator displays the current position. The operation indicator has 33 scale divisions totally. One full turn of the hand crank moves the operation indicator by one division. To complete one full tap-change operation (from one tap position to the next), turn the hand crank for 33 cycles. Then, the operation indicator will complete one full rotation. The counter shows the total number of tap-change operations that the tap-changer has completed.

3. Remove the hand crank. The CMA7 will return to the electric mode.

After the operation, you must make sure the operation indicator returns to the stop position where the red line is.

8. Inspection, maintenance and disposal

DANGER



Electric shock hazard!

Working on live transformers can result in death or serious injury.

- ✓ De-energize and disconnect the high-voltage and low-voltage terminals of the transformer.
- ✓ Lock out the transformer to prevent accidental re-energization.
- ✓ Be sure to de-energize all components.
- ✓ Ground all terminals of the transformer (including the grounding wire and grounding disconnect). Be sure to short-circuit them.
- ✓ Cover or isolate adjacent live parts.

NOTICE

Risk of CMA7 damage!

Condensation inside the CMA7 may cause damage.

- ✓ Always seal the CMA7.
- ✓ If the shutdown period exceeds two weeks, connect and operate the anti-condensation heater inside the CMA7. If you cannot meet the said requirements, place sufficient desiccant inside the CMA7 enclosure.



Important:

For CMA7 troubleshooting and repair, contact the after-sales service department of Huaming.

8.1. Regular maintenance inspection

Follow the inspection instructions in the table below:

Table 8-1 Inspection instructions

Frequency	Inspection item	Inspection procedure
Annual inspection	Door seals, cable entries, CMA7 enclosure ventilation, and control panel ventilation	Visually check for aging or damage. Replace any damaged or aged components. Perform a ventilation test on the CMA7 and control panel.
Annual inspection	CMA7 enclosure and control panel heater	Use a thermometer to perform a temperature test. Verify whether the heater reaches its specified operating temperature.
Annual inspection	Appearance of mechanical components inside the CMA7	Check gears, linkages, and bearings for wear or looseness.
Annual inspection	Appearance of electrical components inside the CMA7	Visually check circuit components for discoloration, damage, or broken pins.
Annual inspection	Perform a functional test on the CMA7 in coordination with tap-changer inspection.	Refer to 6.2 CMA7 functional test .

For any abnormal conditions, contact the after-sales service department of Huaming.

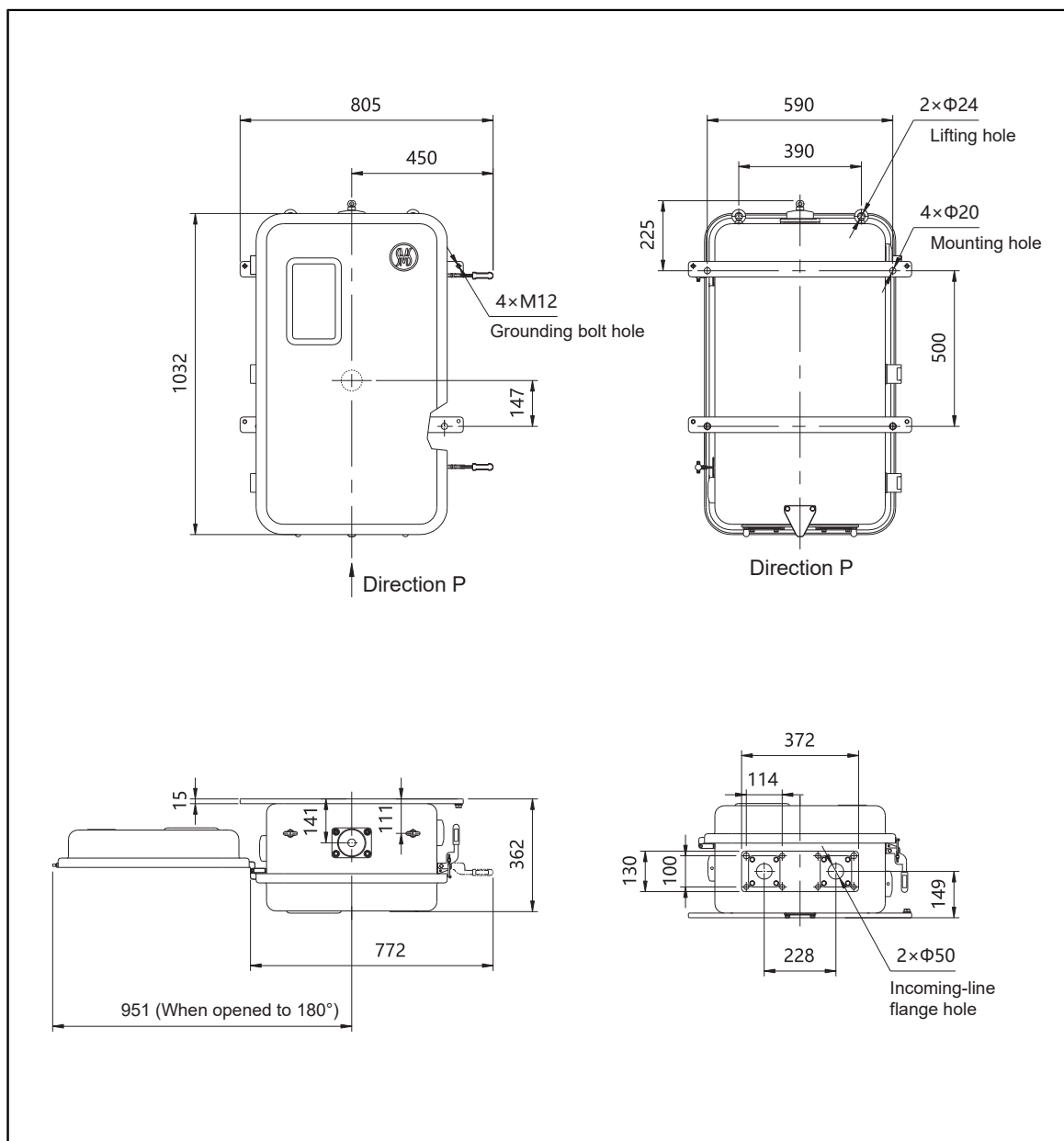
8.2. Disposal

Disposal must obey the applicable national regulations of the country of use. If you have any questions regarding removal or disposal, contact the after-sales service department of Huaming.

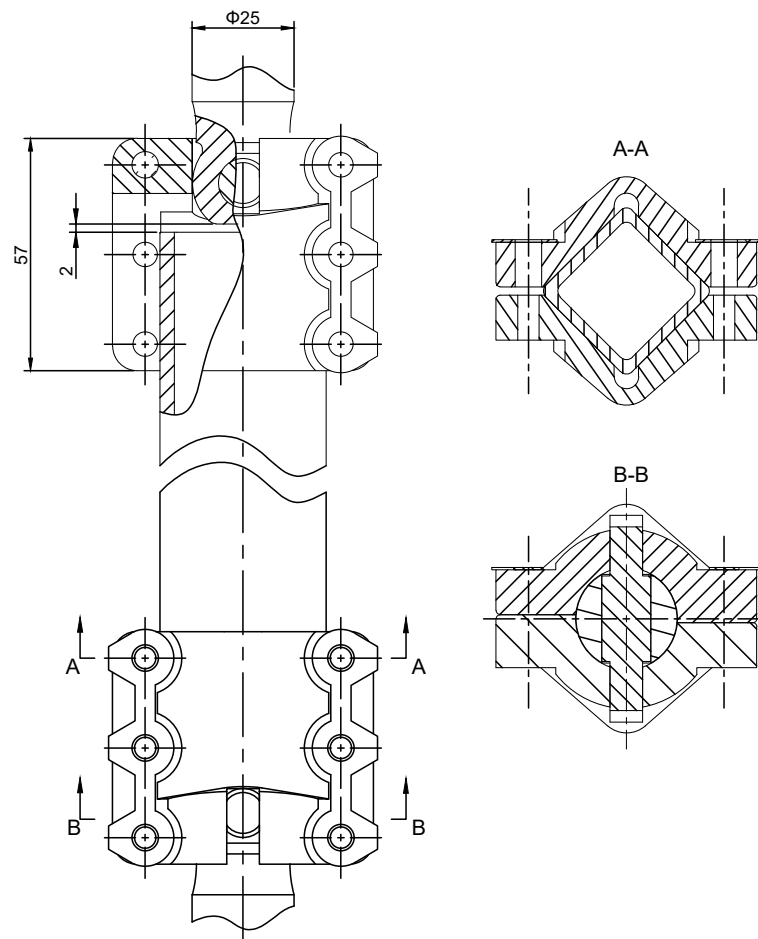
9. Drawings

This section includes the main dimension drawings and electrical drawings of CMA7 Motor Drive Unit.

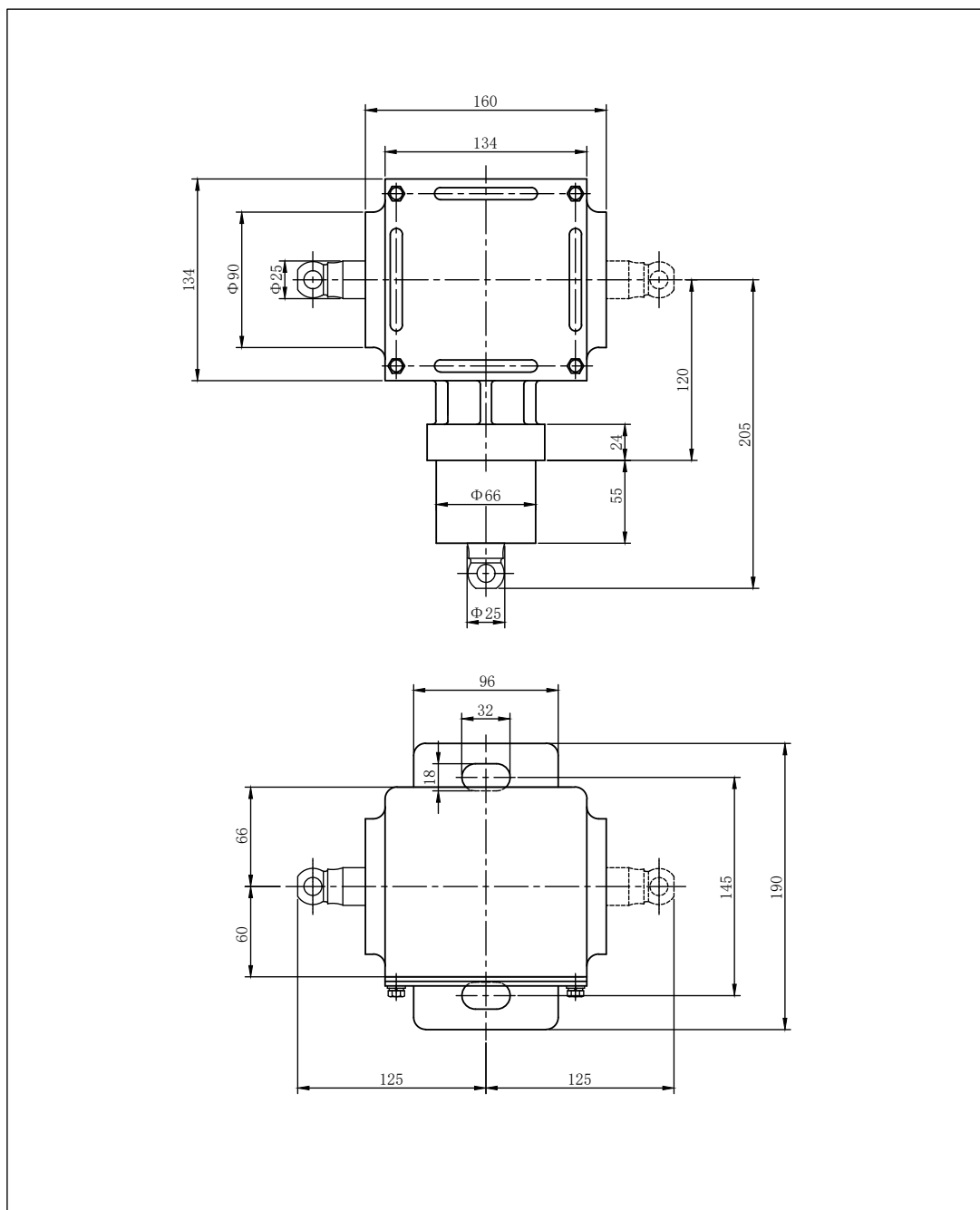
9.1. CMA7 MDU installation dimension drawing



9.2. Horizontal and vertical driving shaft installation



9.3. Bevel gearbox dimensions



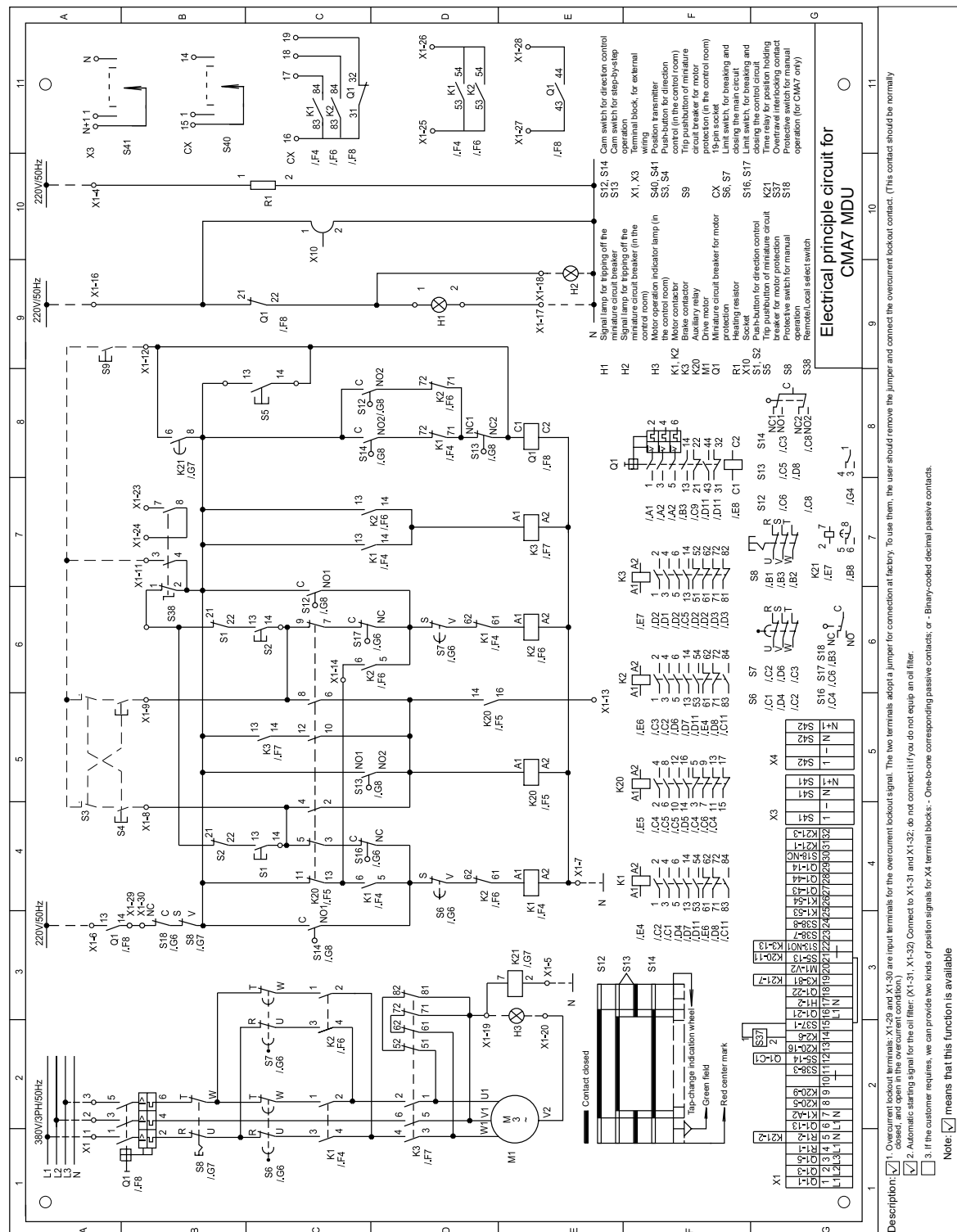
9.4. Designation of Terminal X1 (examples)

Terminal X3 provides one-to-one corresponding position passive contacts. Among them, **X3-N+1** is the common terminal, and **X3-1** to **X3-N** correspond to Positions 1 to N of the tap-changer.

The signals from Terminal X1 vary with the specific model. The table below is only for reference as an example.

Terminals of X1 terminal block	Description
1, 2, 3, 5	Power-in end; voltage of L1, L2 and L3: 380 V/50 Hz; voltage of L1 and N: 220 V/50 Hz
8	Input terminal for remote control "1→N" operation command
9	Input terminal for remote control "N-1" operation command
10, 11	Common terminal for remote control operation command
12	Input terminal for remote control "Stop" command
18	Output terminal for emergency trip signal (outputting 220 V/50 Hz power supply signal)
19, 20	Output terminal for motor operation signal (outputting 220 V/50 Hz power supply signal)
23, 24	Output terminal for "Remote" state signal of "Local/Remote" switch (passive contact)
25, 26	Output terminal for motor operation (passive contact)
27, 28	Output terminal for "Close" state signal of miniature circuit breaker (passive contact)
29, 30	Overcurrent blocking node
31, 32	Automatic starting signal for oil filter

9.5. Electrical principle circuit (example)



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